1) Sunny	Hot	High	Weak	No
2) Sunny	Hot	High	Strong	No
3) Rainy	Mild	High	Strong	No
4) Cloudy	Hot	High	Weak	Yes
5) Rainy	Mild	High	Weak	Yes
6) Rainy	Cool	Normal	Weak	Yes
7) Rainy	Cool	Normal	Strong	No
8) Cloudy	Mild	Normal	Strong	Yes
9) Sunny	Cool	Normal	Weak	No

From the dataset Total rows = 2

prime attribute = Rain

Yes = 1

No = 1

$$\text{Entropy} = -\frac{1}{2}\log_2\left(\frac{1}{2}\right) - \frac{1}{2}\log_2\left(\frac{1}{2}\right)$$

$$= 1$$

Outlook:

	Total	Yes	No	Entropy
Sunny	3	0	3	0
Rainy	3	2	1	0.918
Cloudy	2	2	0	0

Sunny:

Entropy = 0 ∵ contain (0) Yes ~~purest~~ purest value

Rainy:

$$\text{Entropy} = -\frac{2}{3}\log_2\left(\frac{2}{3}\right) - \frac{1}{3}\log_2\left(\frac{1}{3}\right)$$

$$= -0.3899 - 0.528$$

$$= 0.918$$

Cloudy:

Entropy = 0

Weighted Entropy:

$$H(\text{S/outlook}) = \frac{3}{8}(0) + \frac{3}{8}(\quad) + \frac{2}{8}(0)$$

$$= 0.344$$

$$\text{Info gain} = (1) - (0.334)$$

$$= 0.655$$

(2)

Temperature

	Total	Yes	No	Entropy
Hot	2	1	1	1
Mild	3	2	1	
cool	3	1	2	

Hot:

$$\text{Entropy} = -\frac{1}{2}\log_2\left(\frac{1}{2}\right) - \frac{1}{2}\log_2\left(\frac{1}{2}\right)$$

$$= 1$$

Mild:

$$\text{Entropy} = -\frac{2}{3}\log_2\left(\frac{2}{3}\right) - \frac{1}{3}\log_2\left(\frac{1}{3}\right)$$

$$= 0.918$$

cool:

$$\text{Entropy} = -\frac{1}{3}\log_2\left(\frac{1}{3}\right) - \frac{2}{3}\log_2\left(\frac{2}{3}\right)$$

$$= 0.918$$

$$\text{Weighted} = \frac{2}{8}(1) + \frac{3}{8}(0.918) + \frac{3}{8}(0.918)$$

$$= 0.9385$$

$$\text{Info gain} = (1) - (0.9385)$$

$$= 0.0615$$

Humidity:

	Total	Yes	No	Entropy
High	4	2	2	
Normal	4	2	2	

High:

$$\text{Entropy} = -\frac{2}{4}\log_2\left(\frac{2}{4}\right) - \frac{2}{4}\log_2\left(\frac{2}{4}\right)$$

$$= 1$$

Normal:

$$\text{Entropy} = -\frac{2}{4}\log_2\left(\frac{2}{4}\right) - \frac{2}{4}\log_2\left(\frac{2}{4}\right)$$

$$= 1$$

Weighted:

$$= \frac{4}{8}(1) + \frac{4}{8}(1)$$

$$= 1$$

$$\text{Info gain} = (1) - (1)$$

$$= 0$$

Windy:

weak

strong

Weak:

Strong:

Weighted

Info g

we c

Windy:

	Total	Yes	No	Entropy
weak	5	3	2	
strong	3	1	2	

Weak:

$$\text{Entropy} = -\frac{3}{5}\log_2\left(\frac{3}{5}\right) - \frac{2}{5}\log_2\left(\frac{2}{5}\right)$$

$$= 0.2922$$

Strong:

$$\text{Entropy} = -\frac{1}{3}\log_2\left(\frac{1}{3}\right) - \frac{2}{3}\log_2\left(\frac{2}{3}\right)$$

$$= 0.918$$

Weighted:

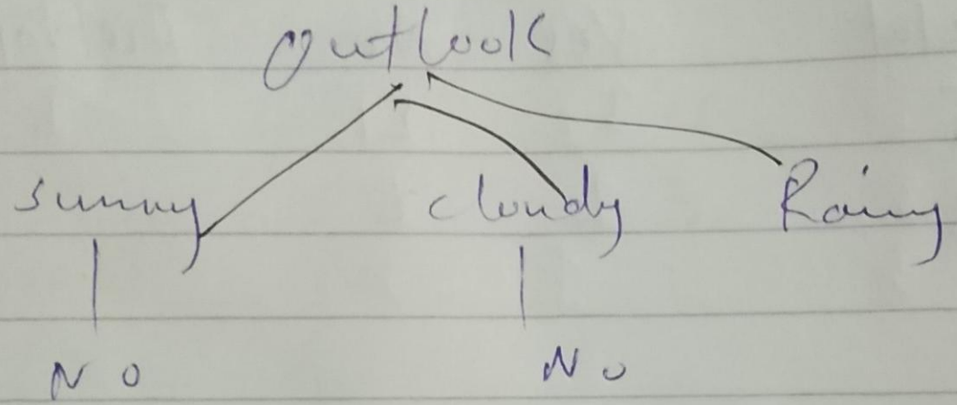
$$= \frac{5}{8}(0.2922) + \frac{3}{8}(0.918)$$

$$= 0.526$$

$$\text{Info gain} = (1) - (0.526)$$

$$= 0.473$$

we choose (Outlook)


Outlook
Sunny
Cloudy
Rainy
No
No